

Written By Doctor Of Pharmacy

Gsc 101 Final Paper All Time

Question No 1:

Types Of Algae?

Ans:

Types of Algae based upon the pigments

☐ **Green algae (chlorophyll, soluble in organic solvents only, such as alcohol), 7000 species**

☐ **Red algae (have “phycobilins” water soluble pigment) 4000 species, cosmeics, gelatin etc**

☐ **Brown algae 1500 species, weeds**

☐ **Golden brown algae (formation of petroleum products)**

☐ **Carotenoids are also algal pigments soluble in organic solvents**

☐ **Euglenoids, both plant and animal like characteristics.**

Question no 2

Biotechnology

Genetic

Answer:

Biotechnology

- **Biotechnology is the use and manipulation of biological processes, organisms and systems to manufacture products to improve the quality of human life.**
- **The term biotechnology was first used in 1970's.**
- **Some important applications**

☐ **Genetic Engineering**

☐ **Fermentation and Microorganisms**

Genetic Engineering

- **It is a process in which a useful gene is taken from one organism and inserted into another organism. The resulting organism is called transgenic.**

- This process is carried out with the help of vectors and enzymes.
- Most common vector is bacterial plasmid.
- Most common enzymes are restriction endonucleases or restriction enzymes.

Question no 3

What is Cell Theory?

Answer:

WHAT IS CELL THEORY?

☐ Cell theory in its modern form states:

- o All living organisms are composed of one or more cells.
- o Cells are the smallest living things, the basic unit of all living organisms.
- o Cells arise only by division in pre-existing cells.

Question no4

Answer:

In females, after first meiotic division (meiosis I), cytoplasm is unequally distributed and one large and other small cell are produced. The small cell is called a polar body. Then both of these cells carry out meiosis II. Polar body divides into 2 polar bodies. The large cell however is divided into one ovum (large) and another polar body. So that meiosis results into 1 ovum and 3 polar bodies.

In males, both meiotic divisions results into equal sized cells called sperms.

Question no 5:

Phase Of Meiosis?

Answer:

Meiosis I

- Meiosis I consist of following phases:

- o Prophase I
- o Metaphase I
- o Anaphase I
- o Telophase I

Question no 6

Preparation Of Hydrogen on Industry?

Answer:

Industrial preparation: On the industrial scale, oxygen is prepared by the following methods.

- **Electrolysis of water:** Electrolysis of water produces hydrogen and oxygen gases.

Liquefaction of air: The major components of air are oxygen and nitrogen. Their boiling points are given below.

b.p. of O₂ = -183°C

b.p. of N₂ = -196°C

So these gases can be separated by liquefaction.

Question no 7:.

Meta and Non Metal Physical and Chemical properties?

Answer:

Metal Physical Properties

- **lustrous (shiny)**
- **good conductors of heat and electricity**
- **high melting point**
- **high density (heavy for their size)**
- **malleable (can be hammered)**
- **ductile (can be drawn into wires)**
- **usually solid at room temperature (an exception is mercury)**
- **opaque as a thin sheet (can't see through metals)**
- **metals are sonorous or make a bell-like sound when struck**

Metal Chemical Properties

- **have 1-3 electrons in the outer shell of each metal atom**
- **corrode easily (e.g., damaged by oxidation such as tarnish or rust)**
- **lose electrons easily**
- **form oxides that are basic**
- **have lower electronegativities**

- are good reducing agents

Nonmetal Physical Properties

- not lustrous (dull appearance)
- poor conductors of heat and electricity
- non-ductile solids
- brittle solids
- may be solids, liquids or gases at room temperature
- transparent as a thin sheet
- nonmetals are not sonorous

Nonmetal Chemical Properties

- usually have 4-8 electrons in their outer shell
- readily gain or share valence electrons
- form oxides that are acidic
- have higher electronegativities
- are good oxidizing agents

Question no 8

Open and Close Circulatory System?

Answer:

: **Closed and open circulatory systems**

: (a) In **closed circulatory systems**, the heart pumps blood through vessels that are separate from the interstitial fluid of the body. ... I

open circulatory system,

the blood is not enclosed in the blood vessels, but is pumped into a cavity called a hemocoel

Question no 9;

What is Tricuspid and Bicuspid Value?

Answer:

Deoxygenated blood from body enters the heart through major veins (vena cava) to right

atrium. When right atrium contract, blood enters the right ventricle.
There is a valve

called tricuspid valve that prevents backflow of blood. On the other side, oxygenated

blood from lungs enters the left atrium and when left atrium contract it enters the left

ventricle. At the junction of left atrium and ventricle another valve is present called

bicuspid valve that prevents backflow of blood. Tricuspid and bicuspid valves are

collectively called atrioventricular valves.

Question no 10

Role of Fatality?

Answer:.

Role in Soil Fertility

- ☐ Some invertebrates are decomposers, e.g., Earthworm
- ☐ They feed upon dead organic matter and increase soil fertility
- ☐ When animals die their bodies become part of soil and decomposed to add nutrients

to soil

Question no 11

Definition?

Arteries

Veins

Capillaries

Answer:

Arteries

are the blood vessels of the body that carry blood away from the heart and to the organs and tissues of the body. The aorta is the largest artery in the body that exits the left ventricle of the heart.

Veins are a type of blood vessel that return deoxygenated blood from your organs back to your heart.

Capillaries

are tiny blood-containing structures that connect arterioles to venules. They are the smallest and most abundant form of a blood vessel in the body.

Question no 12

Difference B/W Types Of Vessels?

Blood vessels

flow blood throughout the body.

Arteries transport blood away from the heart.

Veins return blood back toward the heart.

Capillaries surround body cells and tissues to deliver and absorb oxygen, nutrients, and other substances.

Question no 13

Difference B\w Macromolecules and Macromolecules?

Answer:

a micromolecule refers to a molecule of relatively small size and low molecular weight

as opposed to a macromolecule, which is large and with greater molecular weight. ... In this regard, a protein as a macromolecule is comprised of micro molecule monomers, amino acids linked together by peptide bonds.

Question no 14:

How we test Hypothesis?

How to Test Hypotheses

1. **State the hypotheses.** Every hypothesis test requires the analyst to state a null hypothesis and an alternative hypothesis. ...
2. **Formulate an analysis plan.** The analysis plan describes how to use sample data to accept or reject the null hypothesis. ...
3. **Analyze sample data.** ...
4. **Interpret the results.**

Question no 15

How play algae important role in producer.

Answer:.

Algae belong to the group of plant-like protists. They are autotrophs that fulfill the **role** of **producer** in ecosystems because they make their own food via photosynthesis, like plants.